

Tim Zhang

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SUMMARY

Full-stack software engineer at Goldman Sachs. Led modernization of legacy compliance and banking platforms across two divisions, eliminating \$1.5M+ in annual licensing; built an AI-assisted development platform adopted across ~40 engineers.

WORK EXPERIENCE

Goldman Sachs *Software Engineer II (Associate)*

Salt Lake City, UT | Dec. 2021 - Present

Global Compliance Division

- *REPS (Regulatory Employee Profile Solution) Modernization*
 - Lead full-stack architect on Compliance's largest platform—1,000+ JSPs, 16 backend services, 20-year-old Struts/JSP stack—replacing it with React 19 + TypeScript and eliminating ~\$1M in annual compliance software licensing.
 - Owned end-to-end front-end architecture: selected the stack (Redux Toolkit, React Router, GS-UI design system layered with styled-components, Axios) and established conventions enabling 8–10 engineers across New York, London, and Bengaluru to ship in parallel without merge or architectural conflict.
 - Designed a Core + Feature monorepo using npm workspaces—shared core package owns platform primitives and TypeScript domain types; feature packages build isolated, lazy-loaded modules consuming the core via typed contracts and ship as independent route-level bundles.
 - Defined an incremental NGINX reverse-proxy migration strategy: new React routes added to proxy config as features rebuild while unmigrated paths continue serving from JSP—enabling page-by-page rollout without a big-bang cutover.
 - Built shared front-end primitives (notably a configurable AG Grid wrapper used across modules) to eliminate duplicate grid implementations and standardize sort, filter, pagination, and column-state behavior platform-wide; deployed across multi-environment Kubernetes via Harbor.
- Employee Trading Compliance Lakehouse Pipeline
 - Built end-to-end data ingestion and reporting platform for personal-trading surveillance covering ~30K Goldman Sachs employees, modernizing the data foundation for analytical reporting and future AI/RAG workloads.
 - Designed daily batch ETL ingesting three domains (transactional trading activity, outside-activity disclosures, employee registration records) from a legacy Sybase OLTP database into a Snowflake lakehouse, orchestrated via internal batch scheduler.
 - Executed a TB-scale historical backfill of 20 years of trading and disclosure data into Snowflake, establishing the analytical baseline and a queryable knowledge layer for future RAG-based compliance tooling; incremental loads run daily.
 - Built the downstream reporting layer on an internal data-modeling platform and delivered interactive surveillance dashboards used by Compliance for ongoing oversight.
- Employee Trade Rule Uplift
 - Re-architected the rule storage and lookup layer for personal-trading compliance—modeled hierarchical rules (division → team → employee) as a tree in Elasticsearch with inheritance semantics, replacing a fully-denormalized SQL representation that duplicated divisional rules across thousands of employee rows.
 - Built the indexing pipeline keeping Elasticsearch in sync with the source database, making ES the authoritative read store for rule-evaluation queries; cut rule-lookup screen load times from ~5 seconds to sub-second and offloaded read traffic from the OLTP system.
 - Built a transformer layer that re-flattens the tree structure into the legacy record format consumed by downstream pre-clearance and surveillance systems, preserving backward compatibility without requiring downstream migration.

Global Banking & Markets Division - *Asset Servicing Team*

- *Asset Servicing Web Platform — Appian Replacement (Lead Full-Stack Architect)*
 - Led full-stack architecture replacing Appian (third-party low-code workflow/BPM platform) with a custom React + Spring Boot solution across the Asset Servicing product suite, eliminating ~\$500K in annual third-party licensing.
 - Architected the React + TypeScript front-end with Redux state management and AG Grid for high-density data tables; paired with a Spring Boot + MongoDB backend and a custom work-queue service re-implementing Appian's workflow-engine functionality in-house with full feature parity.
 - Onboarded the Exception Hub: a workflow surface where allocation/holdings mismatches and payment-reconciliation breaks are surfaced to operations users for resolution, with downstream events published to Kafka for consuming systems.
 - Designed the front-end for long-term scalability—data logic isolated in Redux reducers, screen behavior externalized to configuration files, and API access centralized through a single client layer—keeping the platform extensible to new business workflows without architectural rewrites.
 - Built reusable component layers abstracting recurring business UI patterns; GitLab CI/CD pipelines for front-end and back-end; Spring Boot services on Kubernetes; migrated authentication to OAuth 2.0 across front-end and back-end via PingFederate integration.

- *US Equities Corporate Event Pipeline*
 - Built event-driven processing handler for U.S. Equities corporate actions (allocations, payments) on the Asset Servicing platform, supporting near-real-time downstream payment dispatch.
 - Implemented Kafka consumers ingesting corporate-event records from the clearinghouse feed, applying allocation logic, and emitting payment instructions to internal ledger and treasury systems with single-digit-second end-to-end latency.
 - Used Java thread pools and the Executors framework to parallelize event processing across the day's announcement volume, with retry logic and idempotent message handling to ensure data integrity and prevent duplicate payment dispatch on Kafka redelivery.
- *Asset Allocation Exception Handling Service*
 - Built the backend service detecting and persisting processing exceptions across the Asset Servicing corporate-event pipeline (clearinghouse feeds from U.S., London, Hong Kong, and Singapore), feeding ops resolution workflows in the Exception Hub.
 - Implemented a generic break-detection framework over the corporate-event stream, capturing the full exception taxonomy—payment mismatches, payment failures, missed allocation submissions, user-action timeouts, and unresolved stuck-state payments—across multi-region clearing flows.
 - Persisted exceptions in MongoDB as the system of record with versioned audit history: every status transition increments a document version and preserves prior state, providing a full audit trail for compliance review and operational forensics.
 - Published exceptions onto the GBM internal pub/sub bus; downstream Work Queue scheduler consumes them, materializes workflow records, and routes to the appropriate ops users via the Exception Hub.

PERSONAL PROJECTS

- *Autonomous AI Assistant Agent* ([Github](#))
 - Built a ReAct-loop agent (Claude Sonnet) on the Claude ecosystem: thinks, calls tools, observes, re-plans — orchestrated against four custom MCP servers handling data ingestion, semantic ranking, communication, and automated task execution.
 - Used MCP as the tool boundary to keep capabilities composable and the agent stack-agnostic; added a human-in-the-loop approval gate so irreversible actions resume only on explicit user confirmation.
 - Stack: TypeScript, Claude Sonnet, MCP, Playwright, SQLite, node-cron

SKILLS

Programming Languages: Java, Typescript, JavaScript, HTML5/CSS3, SQL

Frameworks & Tools: Spring Boot, React, Docker, Kubernetes, MongoDB, MySQL, Snowflake

AI / Agents: LangChain, LangGraph, Claude API, Claude Code, MCP, ReAct, agentic systems, prompt engineering

EDUCATION

University of Southern California

M.S. in Computer Science; GPA: 3.5/4.0

Los Angeles, CA

Aug. 2019 - Dec. 2021

Pennsylvania State University

B.S. in Information Science & Technology; GPA: 3.8/4.0 - Dean's List

University Park, PA

Sep. 2015 - May. 2019

B.S. in Security Risk Analysis; GPA: 3.7/4.0